

Roxy Denenberg

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EDUCATION

Boston University, College of Engineering

Boston, MA

Electrical Engineering

- Concentrations: in Nanotechnology and Energy Technologies and Sustainability Expected May 2028
- Coursework: Electromagnetism, Intro to Electronics, Modern Physics, Energy system, Circuits, Modern Physics, Linear Algebra, Differential Equations, Digital Logic (Verilog)
- Upcoming (Spring 2027): Signals and Systems, Analog Electronics, VLSI design, Nanotechnology, and Physics of Semiconductors

TECHNICAL SKILLS

- Hardware & Fabrication: Soldering: (SMT and THT), Hot Air Rework, Lab Equipment (Oscilloscope, NanoVNA, Rigol DL3021)
- Software: Altium Designer, Arduino IDE, Jira, Microsoft office, Onshape
- Programming languages: Verilog, C++, Python, MATLAB
- Core Competencies: Signal Integrity, RF, Power Distribution, QA & DCIR Testing, Technical Writing

TECHNICAL WORK EXPERIENCE

ANT61

Sydney, NSW Australia

PCB Electrical Engineering Intern

Jan 2026 – June 2026

- Designed and iteratively revised THREE satellite hardware PCBs using Altium Designer, managing part selection, schematic routing, and final assembly
- Assembled boards with fine pitch components utilizing SMT reflow soldering, THT soldering, and hot-air techniques
- Tested over 500 u.fl cables finding insertion loss across three markers using a Vector Network Analyzer (NanoVNA-H4) to determine integrity
- Performed pre-potting electrical QA testing on hardware and established FFT battery voltage ripple testing as standard practice

Boston University

Boston, MA

Introduction to Engineering Design Teaching Assistant (EK210)

Sept 2026 – Present

- Mentored 30+ students in engineering design principles, assisting with CAD modeling, circuit troubleshooting, and project fabrication

PROJECTS

Iridium Pass Through PCB

March – May 2026

Altium Designer, Jira, THT & SMT Soldering with fine pitch, Hot Plate

- Created in Altium Designer to strict dimensional constraints matching the Iridium 9603 module footprint
- Iterated through board revisions after connector misplacement post-fabrication; performed mechanical fit check prior to V2 fabrication
- SMT soldered fine pitch connectors and decoupling capacitors; THT soldered header connectors for logic analysis and break out purposes

Skittle Color Sorter

Oct – Dec 2025

Part selection, Tinkercad Schematic, Onshape, Machining (Dremel, Drill Press, Lazer cutter)

- Designed schematics in Tinkercad and led component selection, calculating power requirements to accommodate varying motor loads
- Engineered a mechanical structure in Onshape, incorporating a rotating disk mechanism for a color-based candy sorting system
- Collaborated with team to fabricate project using shop machinery and managed cable routes to minimize exposed wiring

Temperature Monitor

Feb – Apr 2025

Wiring components, Tinkercad Schematic, Arduino IDE, CAD (Onshape), THT Soldering

- Developed enclosure with cutouts for an LCD screen, power on button, buzzer, and LEDs as well as battery holder in Onshape featuring a force holding mechanism for ease of replacement
- Wired components together, soldered connections, and used a color-coded wiring system for easy identification
- Programmed Arduino-based monitoring system to trigger audible alerts via buzzer when values exceed thresholds (80°F–90°F)

Medical Light-Up Boot | Lutron Light Challenge

Feb 2025

Wiring components, Arduino IDE, CAD (Onshape)

- Developed custom shell in Onshape, featuring slider rail mechanisms for part assembly and simplified manufacturing
- Coded an ultrasonic sensor via Arduino IDE to automatically activate custom LED lighting sequences upon 3-inch proximity to ground

ADDITIONAL WORK EXPERIENCE

Starbucks

San Diego, CA & Boston, MA

Barista

Jun 2024 – Jan 2025 | Sept 2025 – Present

- Balanced 13+ weekly work hours alongside full-time engineering coursework while keeping dependable and punctual attendance
- Trained new baristas on drink routines, workflow, and customer service expectations